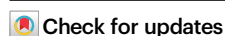


Efficient Pt₁Ni single-atom alloy catalyst for hydrogen-free catalytic fractionation of lignocellulose

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The production of monophenols from lignocellulose with cellulose intact without exogenous hydrogen via self-hydrogen supplied fractionation is promising, yet requires high metal loadings and yields saturated products. Herein, we report a single-atom alloyed Pt₁Ni catalyst that achieves 50.9 wt% yield of phenolic monomers from birch sawdust with about 50% selectivity to valuable propenyl side-chained products under mild conditions (140 °C, 1 atm N₂),

for maximizing Pt utilization and producing unsaturated chemicals from biomass under hydrogen-free conditions, advancing sustainable biorefining.

Given with the ongoing concerns about global warming due to the overuse of fossil energy, more and more attention is being paid to the green, sustainable and renewable biomass resources. Lignocellulosic biomass containing cellulose (30–50%), hemicellulose (20–35%), and lignin (15–35%), represents the most abundant renewable carbon resource on the earth^{1–5}. Efficient and rational fractionation, depolymerization, and upgrading of lignocellulose for the production of high-value chemicals, fuels, and functional materials are of great importance^{6–9}. Traditionally, fractionation of lignocellulose is a carbohydrate-oriented process, aiming to valorize (hemi)cellulose, whereas the valorization of lignin is ignored. While lignin, as one of the three major components of lignocellulose and the largest source of aromatics in nature (mainly composed of monolignols (*p*-coumaryl alcohol, coniferyl alcohol and sinapyl

alcohol¹⁰), its valorization with (hemi)cellulose is paramount for the full utilization of lignocellulose.

The lignin-first biorefining has emerged as a pivotal strategy. This overarching strategy facilitates the active stabilization of reactive lignin intermediates during biomass fractionation to prevent condensation reactions. Broadly, the lignin-first encompasses various stabilization methods including reductive catalytic fractionation (RCF)^{11,12}, oxidative catalytic fractionation (OCF)^{13,14}, protection-group approaches¹⁵. Among these methods, RCF is widely used in biomass fractionation due to the high lignin phenolic monomers yield and good (hemi)cellulose retention^{16,17}. The product selectivity of RCF is highly tunable based on the catalytic system^{18–24}, typically favoring propyl-, propanol-, or propenyl-substituted phenols. Of these, the propenyl variants have garnered significant attention due to their unique double

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bonds, which enable functionalization into various high-value chemicals²⁵. Samec et al. achieved 49 wt% yield of 2,6-dimethoxy-4-(prop-1-enyl)-phenol from birch wood using ethanol/water as solvent over 5 wt% Pd/C catalyst²⁶. In addition, a choline chloride/ethylene glycol-based system was developed for hydrogen transfer reduction catalytic fractionation, yielding 36.2% propenyl phenols with 87.6% selectivity, but resulting in partial degradation of carbohydrates²⁷. Besides, catalysts with low loading and high dispersion also demonstrate outstanding performance in RCF. Song and coworkers reported the atomic dispersed ruthenium catalyst (Ru/ZnO/C) and palladium catalyst (Pd/ZnO/C) for acquiring propenyl-catechol (selectivity of 77%) and propenyl-phenols (selectivity of 62%), respectively^{28,29}. Despite of the progress mentioned above, these processes were carried out at high temperature in organic solvent and consumed exogenous hydrogen (H_2 or alcohols).

Recently, our group developed an innovative self-hydrogen supplied catalytic fractionation (SCF) strategy by using structural hydrogens in hemicellulose as a hydrogen source. Over 2Pt/NiAl₂O₄ catalyst, the fractionation and depolymerization of lignocellulose to phenolic monomers (yield of 46.6 wt%) was realized in water with cellulose intact^{30,31}. Despite obvious advantages, such as operating without exogenous hydrogen, using only water as a solvent, and mild reaction conditions (140 °C), the SCF process still produces products with saturated side chains and requires a high Pt loading (2 wt%). Given that propenyl- or propanol-terminated products are more valuable, we questioned whether unsaturated side groups could be retained through the design of single-atom or single-atom alloy (SAA) catalysts. It well studied that SAA catalysts can offer unique active sites with high activity and improved sintering resistance^{32–34}, so a tailored SAA catalyst for SCF process to enable unsaturated product formation under mild conditions was designed.

Herein, we reported a single-atom alloy (SAA) catalyst comprised of isolated Pt atoms stabilized by Ni atoms on NiAl₂O₄ spinel support for SCF of birch wood. The Pt₁Ni SAA catalyst was prepared by co-impregnation followed by H_2 reduction, achieved nearly theoretical maximum yield of phenolic monomers (50.9 wt%), corresponding to turnover numbers (TON) of 270.5 mol_{phenols} mol⁻¹_{Pt}. Surprisingly, propenyl side-chained syringol (selectivity 51.8%) co-existed with the propyl side-chained syringol at 140 °C for 12 h, the former has more functionality and can be used for bioactive molecules, pharmaceuticals, and materials synthesis³⁵. The SCF over Pt₁Ni SAA catalyst resulted in a cellulose-rich pulp with high glucan retention (90%) due

this manuscript, we noted that Liao's work³⁶ reported a 79.6% selectivity for alkene-functionalized side-chain products, which was achieved under N_2 at 235 °C with methanol serving as solvent and hydrogen source via a different reaction pathway. Both studies efficiently produced unsaturated side-chain products, offering new strategies for the valorization of lignocellulose. The following work would be concentrated on designing more efficient catalyst to achieve high yield of alkene-functionalized side-chain products via SCF.

Results

Characterizations of the structure of the single-atom alloyed Pt₁Ni catalyst

0.2Pt5Ni/NiAl₂O₄ and 2Pt5Ni/NiAl₂O₄ catalysts were prepared by co-impregnation followed by H_2 reduction. As comparison, 2Pt/NiAl₂O₄ and

5Ni/NiAl₂O₄ catalysts were also prepared with the same method. The former (monometallic Pt loaded on spinel) was selected as a high-performance benchmark based on our previous work³⁰, and the latter (monometallic Ni loaded on spinel) was chosen to determine the intrinsic baseline activity of pure Ni catalyst. The phase structures of these catalysts were characterized by X-ray diffraction (Fig. S1) and only the diffraction peaks ($2\theta = 19.1^\circ$, 31.4° , 37.0° , 45.0° , 59.7° , and 65.5°) corresponding to the (111), (220), (311), (400), (511), and (440) planes of NiAl₂O₄ (PDF #10-0339) were observed. No diffraction peaks of Pt and Ni were detected in all of the catalysts, probably due to the low loading or small particle size. N_2 sorption showed that the surface area changed less, all in the range of 61.6–93.0 m² g⁻¹. ICP-AES measurement gave the element composition of Pt (0.095, 0.97, and 1.7 wt%) and Ni (4.3, 4.7 wt%) in all samples and CO pulse chemisorption clearly revealed the good dispersion of Pt (58.5–95.5%). The physicochemical parameters of various samples were summarized in Table S1. The H_2 -TPR profiles of various catalysts were shown in Fig. S2. It shows that monometallic 5Ni/NiAl₂O₄ catalyst has a single reduction peak at 416 °C, attributed to the reduction of NiO species³⁷. For 2Pt/NiAl₂O₄ catalyst, the reduction peak at 193 °C can be assigned to the reduction of PtO_x and the peak at 401 °C is related to the co-reduction of PtO_x and NiO_x species in NiAl₂O₄³⁸. For PtNi/NiAl₂O₄ catalysts with two different Pt loadings, the reduction peaks of NiO species moved to lower temperature because the doping of Pt makes it easier for NiO to be reduced. In addition, the reduction peaks of PtO_x were not detected for the 0.2Pt5Ni catalyst. This is likely due to the low loading of Pt (0.2 wt%), which falls below the detection limit of the TCD. Specifically, the H_2 -TPR profiles reveal that Pt facilitates NiO reduction via hydrogen spillover, where dissociated hydrogen on Pt reduces adjacent NiO species. This confirms the structural intimacy of Pt and Ni, supporting potential alloy formation and enhancing overall catalyst reducibility.

To identify the position of Pt in 0.2Pt5Ni/NiAl₂O₄ catalyst, aberration-correction high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and energy-dispersive spectroscopy (EDS) elemental mapping were conducted. The EDS elemental mappings illustrate a high dispersion and close contact of Pt on Ni nanoparticles (Fig. 1a, b). HAADF-STEM image displays the lattice spacing of -0.181 nm in 0.2Pt5Ni catalyst (Fig. 1c). This lattice spacing corresponds to that of Ni (200) plane, but larger than that of pure Ni (200) plane of 0.176 nm, indicating the occurrence of lattice expansion. Enlarged STEM image shows that number of atom-sized bright spots (highlighted by the red cycles) distributed on the surface of Ni NPs and these atoms are attributed to individual Pt atoms. Meanwhile, no Pt clusters or nanoparticles were observed (Fig. 1d), which demonstrated the formation of Pt₁Ni single-atom alloy. In comparison, the HAADF-STEM image of 2Pt5Ni/NiAl₂O₄ catalyst was also presented (Figure S3) and both Pt single atoms and Pt clusters are detected on Ni NPs, indicating an aggregation of Pt atoms due to the increase in loading. Meanwhile, the 2 wt% Pt/NiAl₂O₄ was synthesized as a control sample, in which Pt clusters were uniformly distributed on the surface of NiAl₂O₄ support with an average particle size of 1.4 nm (Fig. S4).

To further explore the structural details of 0.2Pt5Ni catalyst, in situ DRIFTS spectra of CO adsorption was recorded. As comparison, 2Pt/NiAl₂O₄, 5Ni/NiAl₂O₄ and 2Pt5Ni/NiAl₂O₄ catalysts were also investigated (Fig. 2a). CO adsorption on 2Pt/NiAl₂O₄ catalyst generates dominant peak at 2066 cm⁻¹, which is assigned to the linearly absorbed CO at Pt clusters^{38,39}. For monometallic 5Ni/NiAl₂O₄ catalysts, almost no CO adsorption is detected. In the case of 0.2Pt5Ni catalyst, the observed peak at 2034 cm⁻¹ is assigned to linearly absorbed CO at Pt single atom, the red shift of the linearly absorbed CO peak indicates an enhanced metallicity of Pt species in PtNi catalyst. This would come from the electron transfer from Ni to Pt, which increases the d- π electron feedback from Pt to CO 2 π^* antibonding orbital^{40,41}. Interestingly, a broad peak at 1835 cm⁻¹ is also observed, which is assigned to bridged-bonded CO at contiguous Ni atoms. For 2Pt5Ni catalyst, in addition to the main peak at

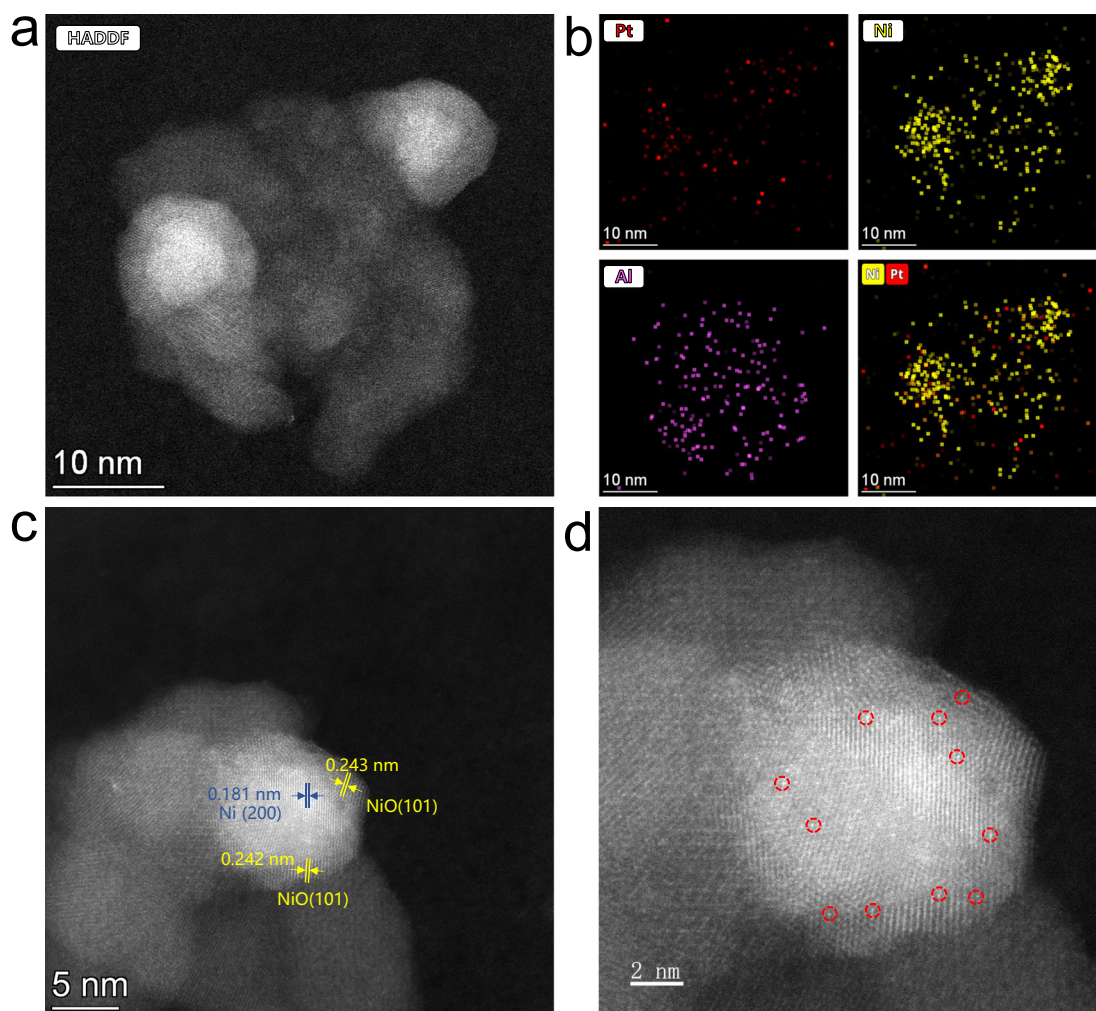


Fig. 1 | Electron microscopy characterization of 0.2Pt5Ni catalyst. **a** HAADF-STEM image, **b** corresponding EDS-mapping images of Pt, Ni, Al, and O elements, **c** AC-HAADF-STEM image with corresponding lattice fringe, **d** enlarged STEM image of (c).

2066 cm^{-1} , an obvious shoulder centered at $\sim 2034 \text{ cm}^{-1}$ is observed, indicating that the catalyst contains a mixture of Pt single atom and Pt cluster, consistent with the STEM observations.

To further study the electronic and geometric structure of PtNi catalysts, the Pt L_3 -edge X-ray absorption near-edge (XANES) spectra were conducted with Pt foil and PtO_2 as reference. As shown in Fig. 2b, the 0.2Pt5Ni and 2Pt5Ni catalysts have lower intensity of “white line” than that of 2Pt/ NiAl_2O_4 catalyst, indicating a more reduced state of Pt species of PtNi catalyst, in accordance with CO-DRIFTS results. The Fourier transforms of the extended X-ray absorption fine spectrum (EXAFS) in the R space was performed to investigate the detailed coordination structure of Pt (Fig. 2c). The 0.2Pt5Ni catalyst exhibits a unique peak located at $\sim 2.1 \text{ \AA}$, which can be assigned to the Pt-Ni coordination^{41,42}. The EXAFS data-fitting results (Table S2 and Fig. S5) show that the coordination number (CN) of Pt-Ni path is 6.6, while Pt-Pt coordination is absent in 0.2Pt5Ni catalyst. This verifies that predominant Pt atoms are atomically dispersed and surrounded by Ni to form single-atom alloy. For 2Pt5Ni and 2Pt/ NiAl_2O_4 catalysts, the Pt-Pt coordination number are 2.5 and 3.4, respectively, indicating the Pt atoms aggregate gradually upon increasing Pt loading. The k^2 -weighted wavelet transforms (WT) for the Pt L_3 -edge XAFS signals were further employed to study coordination environment of Pt species. As shown in Fig. 2d–f, the WT contour plots of 0.2Pt5Ni catalyst display only one Pt-Ni coordination intensity maximum at $2.1 \text{ \AA}$, $7.2 \text{ \AA}^{-1}$, without other scattering paths compared with the WT plots of Pt foil and

PtO_2 , demonstrating the formation of SAA. In brief, the results of AC-HAADF-STEM, CO-DRIFTS as well as XAFS experiments confirm the successful preparation of Pt_1Ni SAA catalyst.

Performance of PtNi/ NiAl_2O_4 in the self-hydrogen supplied catalytic fractionation (SCF)

Birch wood, a typical hardwood substrate containing 43.6 wt% of cellulose, 18.4 wt% of hemicellulose and 20.1 wt% of lignin, was initially used for SCF. The reaction was performed at 140°C and 1 atm N_2 in water with the substrate to catalyst ratio at 0.5/0.2. Firstly, we evaluated the performance of pristine NiAl_2O_4 support and found that almost no product was detected, indicating that the pure carrier is basically inactive (Table 1). The 5Ni/ NiAl_2O_4 catalyst exhibited no measurable activity, which is consistent with the finding that its surface is primarily composed of NiO species, confirming the inactivity of NiO in this reaction. In contrast to monometallic catalysts, all PtNi catalysts showed significantly enhanced catalytic activity (Table 1). As shown in Table 1, 0.2Pt5Ni catalyst achieved the yield of phenolic monomers (38.6 wt%) with 51.8% selectivity to 4-propenyl guaiacol/syringol (Pe-G/S) at 140°C for 12 h (Fig. S6), corresponding to TON as $216.6 \text{ mol}_{\text{phenols}} \text{ mol}^{-1}_{\text{Pt}}$ (The TON was calculated based on the total number of moles of Pt in 0.2Pt5Ni catalyst). When the reaction time was prolonged to 24 h at 140°C , the phenolic monomers yield further increased to 50.9 wt%, approaching the theoretical maximum yield of phenolic monomers¹⁴, with the TON reached $270.5 \text{ mol}_{\text{phenols}} \text{ mol}^{-1}_{\text{Pt}}$, larger than most results of reported catalytic

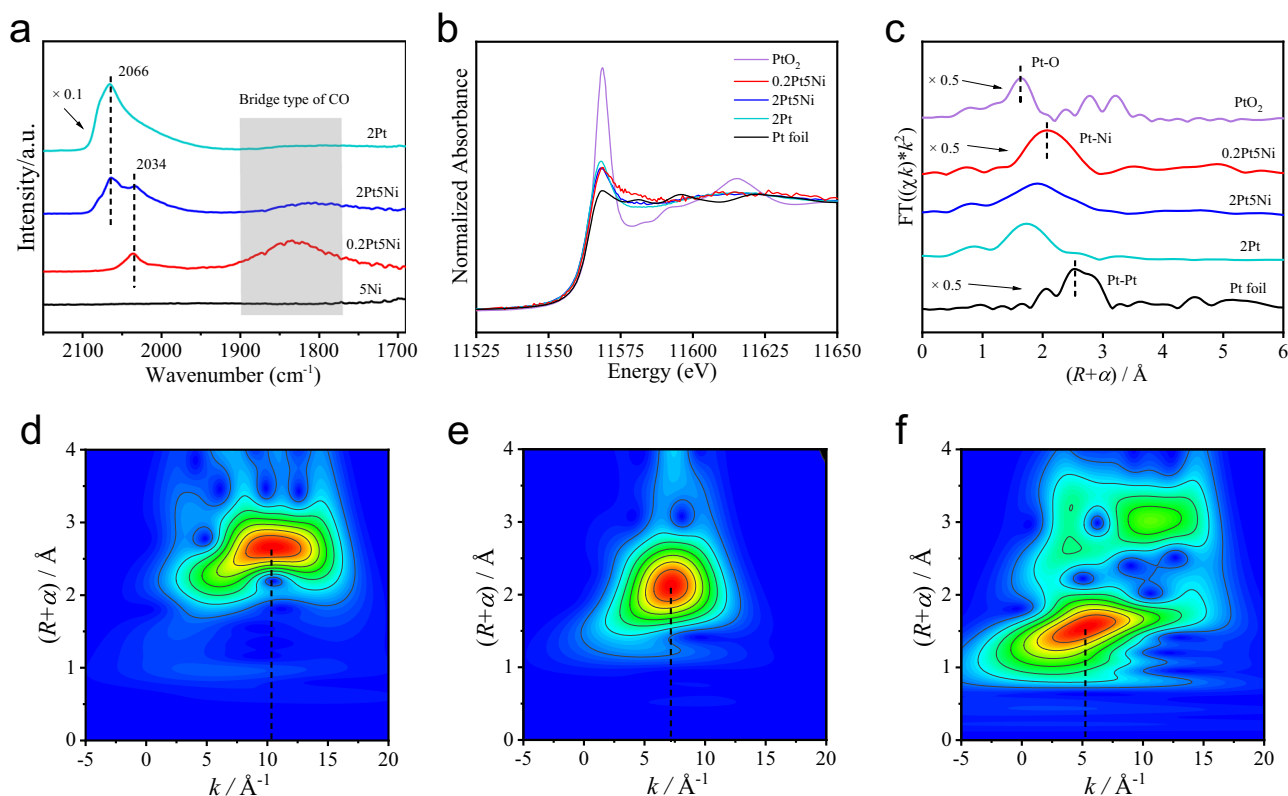


Fig. 2 | Fine-structure characterizations of PtNi catalysts. **a** In situ CO-DRIFTS spectra of 2Pt/NiAl₂O₄, 5Ni/NiAl₂O₄, 2Pt5Ni, and 0.2Pt5Ni catalysts. **b** Pt L₃-edge XANES spectra and **c** EXAFS Fourier-transform spectra of various samples. **k**²-

weighted wavelet transforms for the Pt L₃-edge XAFS signals of **d** Pt foil, **e** 0.2Pt5Ni sample, and **f** PtO₂ based on Morlet wavelets (The data are not phase-corrected).

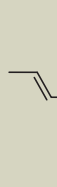
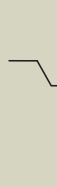
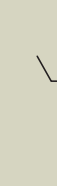
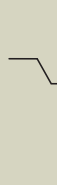
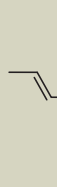
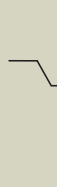
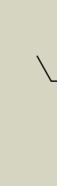
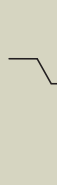
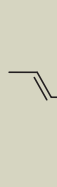
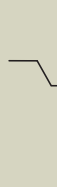
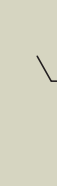
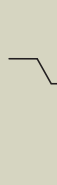
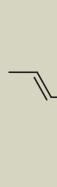
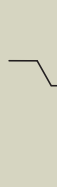
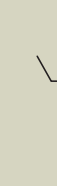
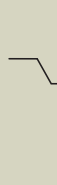
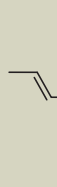
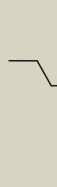
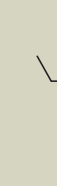
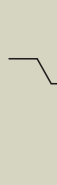
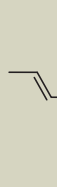
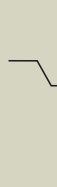
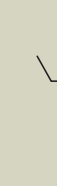
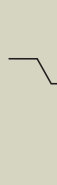
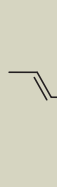
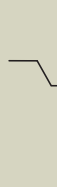
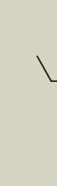
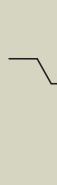
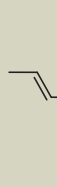
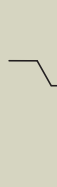
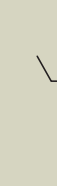
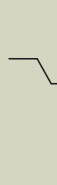
systems (Fig. S7), indicating the superiority of single atom alloyed Pt₁Ni catalyst. In contrast, the 0.2Pt/NiAl₂O₄ catalyst (Table 1, entry 7) was inert, indicating that Pt-Ni interactions are crucial for the SCF process (Tables S3 and S4). This conclusion was further supported by a control experiment using a mechanical mixture of the monometallic catalysts (Table 1, entry 8). Collectively, the characterization and control experiments indicated that the high activity of the Pt₁Ni catalyst originates from the single-atom alloy structure, which effectively stabilizes metallic Ni species. Interestingly, among all monomers obtained over Pt₁Ni catalyst, 4-propenyl syringol (Pr-S, 19.9 wt%) was always co-existed with 4-propyl syringol (Pr-S, 23.5 wt%) as major products, with 4-propenyl guaiacol (Pe-G, 0.9 wt%), 4-propyl guaiacol (Pr-G, 3.2 wt%) and 4-ethyl syringol (Er-S, 3.4 wt%) as the minors (corresponding birch lignin monomers concentrations are shown in (Table S5)). The structural features and molecular weight distribution of the obtained lignin oil were characterized by 2D HSQC NMR and GPC, respectively (Figs. S8 and S9). In addition, the glucan in the pulp showed the good retention (90%) based on the 43.6 wt% of cellulose in the birch and the pulp also showed the 70% of the crystallinity index (CrI) (Fig. S10). 2Pt and 2Pt5Ni catalyst also obtained good phenolic monomers yield of 36.2 wt% and 48.5 wt%, respectively. But almost no 4-propenyl guaiacol(syringol) were detected in the obtained products, and the TONs were calculated as 34.9 and 30.5 mol_{phenols} mol⁻¹_{Pt}, respectively, which were far lower than that from 0.2Pt5Ni catalyst, the latter indeed realize the maximization of Pt utilization efficiency.

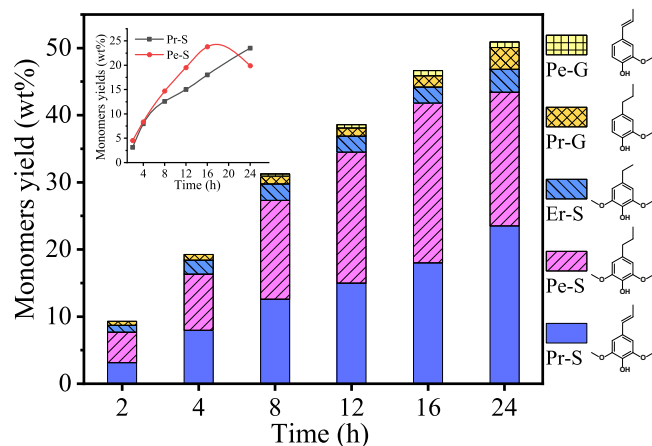
Furthermore, we conducted SCF of birch wood over different PtNi catalysts based on same Pt usage (by adjusting catalyst dosage, Table S6), and found only 2.2 wt% phenolic monomers yield were achieved over 2Pt5Ni catalyst, but 0.2Pt5Ni catalyst still remain high performance (38.6 wt% phenolic monomers yield), further confirming the high efficiency of 0.2Pt5Ni alloy catalyst. To better elucidate the unique role of the NiAl₂O₄ support in the reaction, we synthesized a

0.2Pt5Ni/Al₂O₃ catalyst using Al₂O₃ as the support for comparison. The results showed that NiAl₂O₄ support exhibited superior performance in the selective depolymerization of lignin. This is driven by abundant oxygen vacancies, promoted hydrogen spillover, and robust metal-support interactions (Tables S7 and S8, Fig. S11–S13). Notably, the superior hydrothermal stability of the NiAl₂O₄ support relative to Al₂O₃ support plays an indispensable role in this process. In addition, various oxygen affinity transition metals (e.g., Nb, Mo, W) were also screened for SCF of birch wood (Table S9). Results indicate that the 0.2Pt5W and 0.2Pt5Mo catalysts yield only 14.7% and 8.9% phenolic monomers, respectively. Notably, no phenolic monomers were detected on 0.2Pt5Nb catalyst. The reason for the above results may be that Nb, Mo, and W are difficult to reduce and cannot form alloy structures with metal Pt, further demonstrating the crucial role of Ni in self-hydrogen supplied fractionation.

Above results showed that over 0.2Pt5Ni catalyst, 4-propenyl syringol (Pe-S) and 4-propyl syringol (Pr-S) are all co-existed as main products, which was different from traditional RCF and SCF strategy over 2% Pt/NiAl₂O₄ catalyst. In the latter processes, 4-propenyl syringol was less detected although it was more useful due to the existence of the active C=C bond in C₃ tail chain. In order to investigate the reaction pathway, the reaction time profile of SCF was conducted and the results were presented in Fig. 3. It is shown that in the first 16 h, the yields of Pr-S and Pe-S both increased with the prolonging of time, indicating probably two reaction pathway co-existed, different from that over 2%Pt/NiAl₂O₄ as we reported before (Fig. S14)³⁰. With the further extension of the reaction time to 24 h, the yield of Pe-S decreased and the yield of Pr-S further increased. A plausible explanation is that as the reaction proceeds, most of the lignin is removed (delignification up to 94%) and no new monomers are produced, hence the Pe-S is gradually hydrogenated to Pr-S in the presence of hydrogen. The hydrogen source origins from hemi-cellulose as we described recently⁴¹.

Table 1 | SCF of birch wood over various PtNi catalysts.^a

Entry	Sample	Phenolic monomers yield (wt%)					Sel. to Pe-G/S (%)					TON		
		Lignocellulose												
		Pe-G	Pr-G	Er-S	Pr-S	Pe-S	Pe-G	Pr-G	Er-S	Pr-S	Pe-S			
1	0.2Pt5Ni	0.5	1.2	2.4	15.0	19.5	51.8	38.6	216.6					
2	0.2Pt5Ni ^b	0.9	3.2	3.4	23.5	19.9	40.9	50.9	270.5					
3	2Pt5Ni	-	2.1	16.6	28.4	1.4	2.9	48.5	30.5					
4	2Pt	-	2.0	6.1	25.3	2.8	7.7	36.2	34.9					
5	5Ni	-	-	-	-	-	-	-	-					
6	NiAl ₂ O ₄	-	-	-	-	-	-	-	-					
7	0.2Pt	-	-	-	-	-	-	-	-					
8	0.2Pt+5Ni ^c	-	0.1	0.2	1.5	3.3	-	5.1	-					

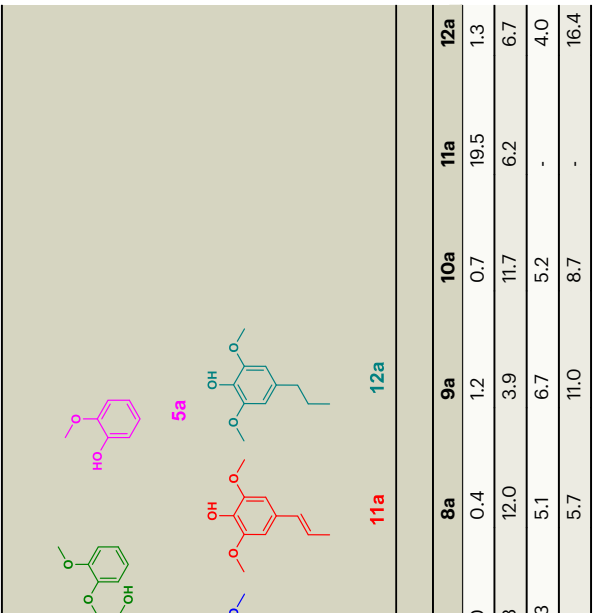
^aReaction conditions: birch wood (0.5 g), catalyst (0.2 g), H₂O (10 mL), N₂ 1 atm, 140 °C, 12 h.^bReaction time: 24 h.^cThe reaction was carried out with 0.2Pt/NiAl₂O₄ (0.1 g) and 5Ni/NiAl₂O₄ (0.1 g).**Fig. 3 |** Time profile of birch SCF reaction over 0.2Pt5Ni catalyst.

Furthermore, the reusability of the catalyst was investigated. The total phenolic monomers yield of 30.5 wt% was achieved with 47.5% selectivity of 4-propenyl guaiacol/syringol (Pe-G/S) at 140 °C for 12 h in the third run (Table S10), a slight decrease yield compared to the first run, indicating the catalyst has good stability in SCF process. The above results imply 0.2Pt5Ni catalyst has excellent activity for SCF of birch wood, and can be further extended to other biomass (pine wood and wheat straw), the total phenolic monomers yield of 28.5 wt% with 26% selectivity of 4-propenyl guaiacol/syringol (Pe-G/S) and 10.8 wt% with 40.7% selectivity of 4-propenyl guaiacol/syringol (Pe-G/S) for pine wood and wheat straw, respectively (Table S11). In addition, the obtained monophenols (lignin oil) could further convert to 4-alkylphenols by demethoxylation in the absence of exogenous hydrogen over RuNi/NiAl₂O₄ catalyst⁴³. Under optimized conditions, 20.5 wt% yield of 4-alkylphenols could be acquired from birch lignin oil (Fig. S15).

Intrinsic catalytic activity and reaction pathways

To further investigate the possible reaction pathways in the SCF process, typical β-O-4 lignin model compound, syringyl-glycerol-β-guaiacyl ether (SG, **1a**), was used to evaluate the reaction pathways over 0.2Pt5Ni/NiAl₂O₄ and 2Pt/NiAl₂O₄ catalysts. To be consistent with the reaction conditions of birch wood, we then added xylan as hydrogen source in the transformation of lignin model compounds. As shown in Table 2, when **1a** was used as the reaction substrate over the 0.2Pt5Ni catalyst for 1 h, the dimeric intermediates, 1-(4-hydroxy-3,5-dimethoxyphenyl)-3-hydroxy-2-(2-methoxyphenoxy)-1-propanone (**2a**), 4-(3-hydroxy-2-(2-methoxyphenoxy)propyl)-2,6-dimethoxyphenol (**3a**), and 4-(3-Hydroxy-2-(2-methoxyphenoxy)propenyl)-2,6-dimethoxyphenol (**4a**) were detected with yields of 2 wt%, 0.8 wt%, and 11 wt%, respectively, indicating there are three potential reaction pathways and the main reaction pathway is the dehydroxylation of C_αOH and formation of C_α=C_β double bond. Importantly, the unsaturated C=C products, derived from the cleavage of **4a**, sinapyl alcohol (**6a**) and its dehydroxylation product Pe-S (**11a**), were detected with yields of 6.1 wt% and 19.5 wt%, respectively. This is consistent with the results of the SCF reaction of birch wood, which primarily yields 4-propenyl products. Additionally, the phenolic monomers with side-chain saturated and oxygen-containing (**7a**, **8a**, **9a**, **10a**, and **12a**) were also detected with yields of 8 wt%, 0.4 wt%, 1.2 wt%, 0.7 wt%, and 1.3 wt%, respectively. With the reaction time extended to 4 hours, the unsaturated C=C products gradually decreased due to hydrogenation and lead to over-reaction, with the yields of **8a** and **10a** increased. In addition, we conducted control experiments on model compound SG using different types of hydrogen sources, such as xylan, isopropanol, and H₂. The consistent reaction pathway was observed, demonstrating that the formation of propenyl-

Table 2 | Reaction pathway of lignin model compound SG at different reaction times



Entry	Catalysts	Time (h)	Conv. (%)	Yield (wt%)											
				2a	3a	4a	5a	6a	7a	8a	9a	10a	11a	12a	
1	0.2Pt5Ni	1	100.0	2.0	0.8	11.0	39.0	6.1	8.0	0.4	1.2	0.7	19.5	1.3	
2	0.2Pt5Ni	4	100.0	4.7	0.3	0.2	43.3	0.3	0.3	12.0	3.9	11.7	6.2	6.7	
3	2Pt	1	100.0	9.2	0.4	-	40.3	-	17.3	5.1	6.7	5.2	-	4.0	
4	2Pt	4	100.0	3.1	0.5	-	44.2	-	1.0	5.7	11.0	8.7	-	16.4	

Reaction Conditions: Xylan (0.05 g), Substrate (0.1 g), catalyst (0.1 g), H₂O (10 mL), N₂ 1 atm, 140 °C.

substituted phenolic products is primarily determined by intrinsic catalytic properties rather than hydrogen sources (Table S12).

In contrast, no dimer intermediate **4a** and the corresponding monomers **6a** and **11a** were produced over the 2Pt catalyst within 1 h of reaction, while a large amount of monomer **7a** (17.3%) was produced, indicating a distinct reaction pathway compared to the 0.2Pt5Ni catalyst. The same pattern was also observed in experiments using hydroxyphenyl-glycerol- β -guaiacyl ether (HG) and guaiacyl-glycerol- β -guaiacyl ether (GG) as the substrate (Tables S13 and S14), suggesting that three reaction pathways co-exist over the 0.2Pt5Ni catalyst, while the 2Pt catalyst has a relatively single reaction pathway (C_{α} -OH dehydrogenation) and is unable to produce the unsaturated dimer intermediates **4a**, **4b**, and **4c**. This might be due to the stronger hydrogen generation and hydrogen migration capabilities of the 2Pt catalyst compared to the 0.2Pt5Ni catalyst (Fig. S16).

(i) C_{α} -OH deH: dehydrogenation of the C_{α} -OH in β -O-4 units to ketone intermediates. Then, the C_{β} -O bond of the intermediate will cleave to monomers, followed by hydrogenolysis and hydrogenation process to obtain 4-propyl-syringol (guaiacol). (ii) C_{α} -OH deOH-I: hydrogenolysis of C_{α} -OH in β -O-4 units, then cleavage of C_{β} -O bond, further to obtain 4-propyl-syringol (guaiacol). (iii) C_{α} -OH deOH-II: hydrogenolysis of C_{α} -OH and dehydrogenation of C_{β} -H to enol ether intermediate, then cleavage of C_{β} -O bond, further to obtain 4-propenyl-syringol (guaiacol). In addition, acid formed in biomass depolymerization may also be involved in the reaction, which also promotes the C_{α} -OH deOH-II pathway and therefore produces more propenyl-ended products.

Discussion

In summary, we developed an innovative Pt₁Ni single-atom alloy (SAA) catalyst for hydrogen-free catalytic fractionation of lignocellulose via self-hydrogen supplied catalytic fractionation (SCF) strategy. The single-atom alloy structure was unambiguously confirmed by multiple characterization techniques, including AC-HAADF-STEM, in situ CO-DRIFTS, and EXAFS. Under mild conditions (140 °C, 24 h), the catalyst achieved a near-theoretical yield of phenolic monomers (50.9 wt%), predominantly propyl/propenyl-substituted syringol derivatives, while preserving cellulose integrity. The system exhibited a high turnover number (TON) of 270.5 molphenols·molPt⁻¹ and demonstrated good

Methods

Catalyst preparation

NiAl₂O₄ spinel was synthesized by a coprecipitation method at a Ni/Al mole ratio of 1:2^{38,45}. A typical procedure is as follows: 40 mmol nickel nitrate hexahydrate and 80 mmol aluminum nitrate nonahydrate were dissolved in 200 ml of ultrapure water under vigorous stirring at room temperature. Aqueous ammonia (25–28 wt%) was added dropwise to the above solution until the pH reached 8 to obtain precipitates. Then, the formed precipitates were filtered and washed with deionized water until the pH of the filtrate reached 7. Finally, the filter cake was dried at 100 °C for 24 h, followed by calcination at 800 °C for 8 h under Ar atmosphere. xPtyNi/NiAl₂O₄ (where x, y represents the loading weight of Pt and Ni, respectively) was prepared by the co-incipient wetness impregnation method with appropriate amounts of aqueous solution of chloroplatinic acid (H₂PtCl₆) and NiCl₂·6H₂O, respectively. The obtained samples were dried at 60 °C for 12 h and then reduced in a 10% H₂/Ar flow at 400 °C for 3 h. To make the Tables/Figures look more concise, we use xPtyNi represent xPtyNi/NiAl₂O₄ catalysts.

Catalyst characterizations

The powder X-ray diffraction (XRD) patterns were recorded on a Rigaku D/max-2550VB/PC diffractometer by using Cu K α radiation (λ = 0.15406 nm). Nitrogen adsorption/desorption isotherms of the catalysts were measured on a Micromeritics ASAP 2020 M sorption analyser at 77 K. The Brunauer-Emmett-Teller (BET) method was used to calculate the surface area. Aberration-corrected-HAADF-STEM images and EDS mappings were collected at 300 kV on a ThermoFisher Scientific Spectra 300 scanning transmission electron microscope equipped with a Super-X EDS detector system. Inductively coupled plasma atomic emission spectrometry (ICP-AES) was used to determine the Pt/Ni loading content by using an Agilent 725ES.

H₂ temperature-programmed reduction (H₂-TPR) experiments were conducted on VDSorb-91i automatic chemisorption system. Typically, 0.1 g of catalyst was injected into the quartz sample tube. Before each run, the catalyst sample was in situ pretreated with Ar at 150 °C for 1 h and cooled down to ambient temperature. Then, the samples were heated to 700 °C with a 10 °C min⁻¹ ramp in 10% H₂/Ar flow.

CO pulse chemisorption tests were operated on VDSorb-91i automatic chemisorption system. Typically, 0.1 g of catalyst was injected into the quartz sample tube. Before each run, the catalyst sample was in situ reduced in flowing 10% H₂/Ar at 400 °C for 1 h and cooled down to ambient temperature with He. The CO pulse chemisorption results were calculated on the basis of the peak areas of the CO pulse injection (220 μ L per pulse, 50% CO/He), which were monitored by a TCD detector (45 °C, 80 mA).

In situ diffuse reflectance infrared Fourier transform (DRIFTS) of CO adsorption was collected with a Nicolet iS50 FTIR spectrometer equipped with an MCT/A detector. Firstly, putting around 20 mg of catalyst into the chamber, the catalyst was in situ reduced in flowing 10% H₂/Ar at 400 °C for 1 h and decreased to 30 °C, followed by collecting the background spectrum. 1% CO/Ar was then introduced into the chamber for 30 min. Next, the chamber was purged with Ar for

another 30 min and the adsorption spectra of CO was recorded. All spectra were collected after subtracting the background with a resolution of 4 cm⁻¹ and an accumulation of 32 scans.

XAS measurements were conducted at the Canadian Light Source for the Pt L₃-edge on the Hard X-ray Micro-Analysis Beamline

(HXMA), beamline. Pt foil was used for energy calibration and both fluorescence and transmission yields were recorded. fluorescence spectra were reported for samples with low Pt loading. Data analysis was performed using a standardized IFEFFIT package (including Athena and Artemis software)⁴⁶.

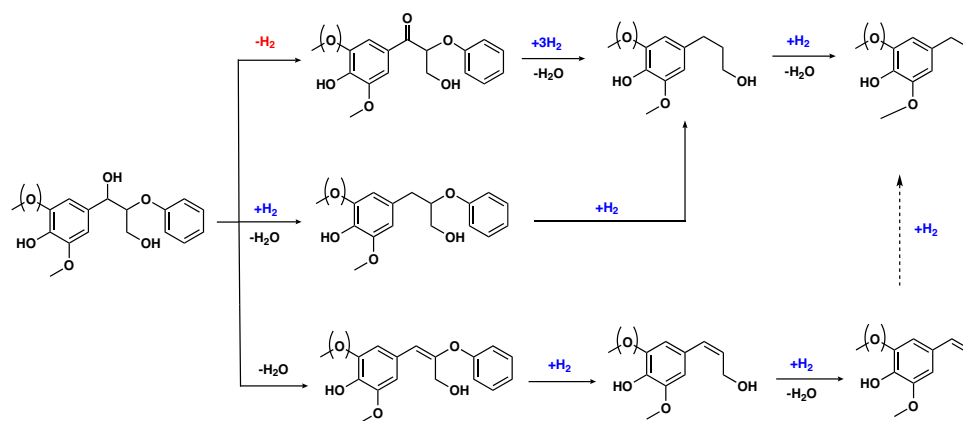


Fig. 5 | Proposed mechanism of lignin depolymerization in SCF.

Catalytic reactions

In a typical SCF reaction, 0.5 g birch wood, 0.2 g xPtyNi/NiAl₂O₄ catalyst and 10 mL H₂O were put into the 50 mL stainless-steel autoclave reactor. The reactor was sealed and purged with N₂ at room temperature, then heated to different temperatures and kept for a certain time at a magnetic stirring speed of 700 rpm. At the end of the reaction, it was cooled down to room temperature in ice water. The soluble fraction and residue were subsequently separated by centrifugation.

Lignin-derived Products Analyses

After SCF reaction, the liquid phase was extracted by ethyl acetate and analyzed by GC-MS (Agilent 7890 A) and quantitatively analyzed by GC (Agilent 7890B) with a flame ionization detector, both equipped with HP-5 capillary columns. Tridecane was used as an internal standard for the quantification of the liquid products. Nitrogen was employed as a carrier gas, and the following operating conditions were adopted: injection temperature 250 °C, column temperature program: 50 °C (hold time 3 min), 10 °C min⁻¹ to 250 °C (hold time 5 min), detection temperature of 280 °C. Response factors for the different products were determined by calibration with commercial standards or through calculations based on the effective carbon number (ECN) theory⁴⁷. All experiments were performed at least three times. The phenolic monomers yields were calculated based on the Klason lignin weight, calculating formula as shown below:

$$\text{Phenolic monomers (wt\%)} = \frac{\text{Mass (total monomers)}}{\text{Mass (Klason lignin)}} \times 100\% \quad (1)$$

$$\text{Mole yields of monomers (\%)} = \frac{\text{Mole (total monomers)}}{\text{Mole (all monomers in lignin)}} \times 100\% \quad (2)$$

The mass of lignin in biomass was measured by Klason method and the mole of monomers in lignin was measured by NBO method⁴⁵.

strength between the functional groups and the surface. To be analogous to the Crystal Orbital Overlap Population (COOP) analysis, -COHP was plotted instead of COHP. Thus, the more negative the -IpCOHP, the stronger the chemical bonding between the adsorbate and the surface.

Data availability

All data generated in this study are provided in the Supplementary Information/Source Data file. Source data are provided with this paper. All data are available from the corresponding author upon request. Source data are provided with this paper.

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Author contributions

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Competing interests

The authors declare no competing interests.

Additional information

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